

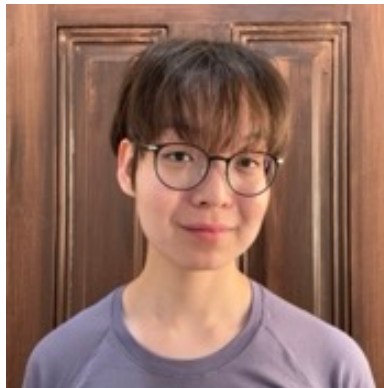
# Statistics 151a: Linear models

UC Berkeley, Spring 2026

## Instructors



**Instructor:** Ryan Giordano (389 Evans) Tues 4pm-5pm (on [Zoom](#)) [rgiordano@berkeley.edu](mailto:rgiordano@berkeley.edu) pronouns: He / him Office: 389 Evans Hall Office hours: Thur 11am-12pm



**GSI:** Danat Dusingbekov (both in 428 Evans Hall) [danat@berkeley.edu](mailto:danat@berkeley.edu) pronouns: He / him Office hours: Wed 1pm-3pm, Mon 12pm-2pm

**! Important**

Please reach out to [Danat](#) if you need to be manually added to BCourses.

## Schedule

Lectures will be held on Monday, Wednesday, and Friday from 9–10am in Davis 534. The first lecture will be on Jan 21st, and the last will be on Friday May 1st.

Labs will be held on Wednesdays from 11 am – 1 pm and 3–5 pm in Evans 340.

([Link to official course calendar.](#))

## Materials

Unelss otherwise noted, the primary materials for the course are the lecture notes, which will be posted to the course website in advance of class. The following textbooks are useful supplementary texts and are all either freely available online or available electronically through the UC Berkeley library:

- [SMT: Statistical Models: Theory and Practice](#) Freedman
- [LME: Linear Models and Extensions](#) Ding
- [ISL: An Introduction to Statistical Learning](#) James, Witten, Hastie, Tibshirani
- [VDS: Veridical Data Science](#) Yu, Barter
- [ETM: Econometric Theory and Methods](#) Davidson, MacKinnon

Other books of interest to the class are

- [FPP: Statistics](#) Freedman, Pisami, Purves
- [RDS: R for Data Science](#), Wickham, Grolemond
- [ROS: Regression and other Stories](#) Gelman, Hill, Vehtari

## Linear algebra review materials

Basic linear algebra is a serious prerequisite for this course. You can find a summary of useful review materials [here](#).

## R programming review materials

This course will be conducting in the R programming language, and basic proficiency is assumed. Here are some useful review materials:

- [Berkeley SCF R bootcamp materials](#)
- [Berkeley D-Lab R fundamentals workshop](#)

## Course Calendar

The following schedule is subject to change.

Table 1: Calendar (tentative)

| Date   | Day       | Note                   | Unit                           | Topic                                | Assignment   |
|--------|-----------|------------------------|--------------------------------|--------------------------------------|--------------|
| Jan 21 | Wednesday | Lecture                | Introduction                   | Class policies                       | Lab 0        |
| Jan 23 | Friday    | Lecture                |                                | Real-world questions                 |              |
| Jan 26 | Monday    | Lecture                |                                | Inference and probability            |              |
| Jan 28 | Wednesday | Lecture                | Unit 1: Multilinear regression | Prediction and uncertainty           | Lab 1        |
| Jan 30 | Friday    | Lecture                |                                | Linear algebra review                | HW 0         |
| Feb 2  | Monday    | Lecture                |                                | Simple linear regression             |              |
| Feb 4  | Wednesday | Lecture                |                                | Multilinear regression               | Lab (review) |
| Feb 6  | Friday    | Lecture                |                                | ———                                  | Quiz 0       |
| Feb 9  | Monday    | Lecture                |                                | Paths to the least squares objective | Lab 2        |
| Feb 11 | Wednesday | Lecture                | Unit 2: Testing and inference  | Transforming the regressors          |              |
| Feb 13 | Friday    | Lecture                |                                | Transforming the response            |              |
| Feb 16 | Monday    | Administrative holiday |                                |                                      |              |
| Feb 18 | Wednesday | Lecture                |                                | Confidence intervals review          | Lab 3        |
| Feb 20 | Friday    | Lecture                |                                | The multivariate normal              | HW 1         |
| Feb 23 | Monday    | Lecture                |                                | Z and T tests                        |              |
| Feb 25 | Wednesday | Lecture                |                                | Model selection and F-tests          | Lab (review) |

| Date   | Day       | Note         | Unit                                        | Topic                                  | Assignment   |
|--------|-----------|--------------|---------------------------------------------|----------------------------------------|--------------|
| Feb 27 | Friday    | Lecture      |                                             | —                                      | Quiz 1       |
| Mar 2  | Monday    | Lecture      |                                             | Asymptotics and heteroskedasticity     |              |
| Mar 4  | Wednesday | Lecture      |                                             | Misspecification                       | Lab 4        |
| Mar 6  | Friday    | Lecture      |                                             | Statistical and practical significance |              |
| Mar 9  | Monday    | Lecture      | Unit 3: Regularization and machine learning | <b>Canceled class</b>                  |              |
| Mar 11 | Wednesday | Lecture      |                                             | Splines and basis expansions           | Lab 5        |
| Mar 13 | Friday    | Lecture      |                                             | Bayesian estimators                    | HW 2         |
| Mar 16 | Monday    | Lecture      |                                             | Ridge regression                       |              |
| Mar 18 | Wednesday | Lecture      |                                             | Lasso regression                       | Lab (review) |
| Mar 20 | Friday    | Lecture      |                                             | —                                      | Quiz 2       |
| Mar 23 | Monday    | Spring break |                                             | —                                      |              |
| Mar 24 | Tuesday   | Spring break |                                             | —                                      |              |
| Mar 25 | Wednesday | Spring break |                                             | —                                      |              |
| Mar 26 | Thursday  | Spring break |                                             | —                                      |              |
| Mar 27 | Friday    | Spring break |                                             | —                                      |              |
| Mar 30 | Monday    | Lecture      | Unit 4: How regression can go wrong         | Outliers and leverage                  |              |
| Apr 1  | Wednesday | Lecture      |                                             | The influence function                 | Lab 6        |
| Apr 3  | Friday    | Lecture      |                                             | Regression to the mean                 | HW 3         |
| Apr 6  | Monday    | Lecture      |                                             | The FWL theorem                        |              |
| Apr 8  | Wednesday | Lecture      |                                             | Omitted variable bias                  | Lab (review) |
| Apr 10 | Friday    | Lecture      |                                             | —                                      | Quiz 3       |
| Apr 13 | Monday    | Lecture      | Unit 5: Generalizations                     | <b>Canceled class</b>                  |              |
| Apr 15 | Wednesday | Lecture      |                                             | Logistic regression                    | Lab 7        |
| Apr 17 | Friday    | Lecture      |                                             | Poisson regression                     | HW 4         |
| Apr 20 | Monday    | Lecture      |                                             | Nonlinear least squares                |              |

| Date   | Day       | Note    | Unit                  | Topic                 | Assignment     |
|--------|-----------|---------|-----------------------|-----------------------|----------------|
| Apr 22 | Wednesday | Lecture |                       | Random effects models | Lab (review)   |
| Apr 24 | Friday    | Lecture |                       | Hierarchical modeling | Quiz 4         |
| Apr 27 | Monday    | Lecture | Unit 6: Special topic | TBD                   |                |
| Apr 29 | Wednesday | Lecture |                       | TBD                   | Lab (projects) |
| May 1  | Friday    | Lecture |                       | TBD                   | HW 5           |
| May 4  | Monday    | Lecture | Final projects        |                       |                |
| May 6  | Wednesday | Lecture |                       |                       | Lab (projects) |
| May 8  | Friday    | Lecture |                       |                       |                |

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